

RUTIKA AVINASH KADAM

Stony Brook, NY 11790 · +1 934-949-8653 · rutikakadam2727@gmail.com · LinkedIn · Portfolio · GitHub

SUMMARY

Data Scientist and AI professional with 4 years of experience building scalable ML and multi-modal GenAI systems focused on rigorous evaluation and measurable impact. Collaborative problem-solver who turns complex challenges into real-world solutions.

PROFESSIONAL EXPERIENCE

Stony Brook Medicine, USA | Data Scientist (Graduate Researcher) **July 2025 – Present**

- Built an end-to-end **multi-modal RAG pipeline** over **6k+ pages** of Department of Public Health medical documents, using Azure Document Intelligence for layout-aware extraction, GPT-4o for figure captioning, and **custom semantic chunking** with adaptive breakpoints.
- Benchmarked FLAT, HNSW, and IVF_FLAT indexes in **Milvus** on 27k+ multi-modal chunks (p50/p95 latency, Recall@5); approximate indexes matched accuracy at **~2.5× lower latency**.
- Evaluated vector search, MMR reranking, and **hybrid Vector+BM25 retrieval with RRF** on a synthetic ground-truth set spanning text, tables, and images, reaching **0.90 Recall@5**.
- Built grounded generation with source citations, scoring **RAGAS** 0.82 faithfulness, 0.82 answer relevancy, **0.93 context precision**, and 0.74 context recall.

Tata Consultancy Services, India | Data Scientist **August 2020 – April 2024**

- Performed **exploratory data analysis** on **1M+ vulnerability records** from Qualys VMDR with the Vulnerability Management team, surfacing trends, anomalies, and threat vectors that guided remediation prioritization.
- Built linear and logistic regression models (scikit-learn) predicting CVSS severity and exploit availability on ~150K findings, achieving **0.74 ROC-AUC** and **22% lower risk exposure** vs. FIFO prioritization in backtests.
- Uncovered recurring remediation patterns in 150K+ records via **semantic clustering** (E5 embeddings + HDBSCAN), then fine-tuned a **DistilBERT classifier** on cluster labels (100% held-out accuracy), laying groundwork for automated ticket routing.
- Built a **Power BI** dashboard with DAX KPIs (CVSS, exploit exposure, MTTR, SLA compliance) that turned alert noise into clear actions for non-technical stakeholders, showing **64% fewer Jira tickets**.
- Achieved **99% patch compliance** by automating enterprise remediation workflows with SCCM and PowerShell, boosting productivity by 25%.

Zensar Technologies, India | Data Intern **May 2019 – July 2019**

- Built **ETL pipelines in Azure Data Factory** with incremental batch loading to ingest multi-channel website traffic and operational data into Azure Data Lake Gen2 for Swiggy-India.
- Developed **Power BI dashboards** for conversion funnels and marketing channel attribution, identifying drivers of traffic and campaign performance and increasing **high-intent traffic by 15%**.

PROJECT EXPERIENCE

Electric Vehicle Battery Capacity Prediction | PyTorch, MLOps, Data Version Control, Azure

- Predicted **EV battery capacity** from everyday charging data, since direct measurement needs conditions present in only 56% of sessions; trained PyTorch models on **111K** charging windows from 30 vehicles with vehicle-wise 5-fold CV.
- Benchmarked LSTM and Transformer models against baselines over 10 tracked experiments; a linear odometer model won (**RMSE 1.10** vs. 1.69/1.71), and a positional-encoding ablation showed the 21-minute window, not model capacity, was the limit.
- Versioned data and models with DVC across two cloud remotes, tracked experiments in MLflow, and deployed a **FastAPI** service via Docker and Azure Container Registry to a public Azure Container Apps endpoint, with **CI running 43 tests** per push.

Airbnb Price Prediction: Seattle | Python, scikit-learn, Bagging and Boosting, NLTK

- Analyzed **3.8K** Seattle Airbnb listings and ran **sentiment analysis** on **84K reviews**; room type and bedroom count drove price most, while review sentiment showed no effect.
- Engineered **19 amenity features** and benchmarked **6 models** (linear, **bagging, boosting**) with leak-free **10-fold CV**, reaching **$R^2 \approx 0.51$** (**RMSE $\approx$ \$62**); boosting overfit.

EDUCATION

Stony Brook University, USA - Master of Science in Data Science **August 2024 – May 2026**

Coursework: Machine Learning, Deep Learning, Natural Language Processing, Large Language Models, Agentic AI, Big Data.

Pune University, India - Bachelor of Engineering in Information Technology **August 2016 – May 2020**

Coursework: Statistics, Discrete Structures, Data Structures & Algorithms, System Design, Cloud Computing.

SKILLS

Programming Languages: Python, R, SAS, SQL (MySQL, PostgreSQL, PL/SQL), C++, HTML, CSS

Data Science: NumPy, Pandas, Seaborn, scikit-learn, TensorFlow, Keras, PyTorch, Hugging Face, LangChain, LangGraph

Data Engineering & Big Data: Azure Data Factory, Data Lake, Databricks, Microsoft Fabric, SSMS, SSIS

Cloud & DevOps: Azure, FastAPI, REST APIs, Docker, Kubernetes, MLflow, Git, GitHub, Postman

Visualization & Analytics: Power BI, Tableau, Microsoft Excel

Security & IT Operations: Qualys VMDR, Microsoft Endpoint Configuration Manager (SCCM), ServiceNow, Jira, Aternity